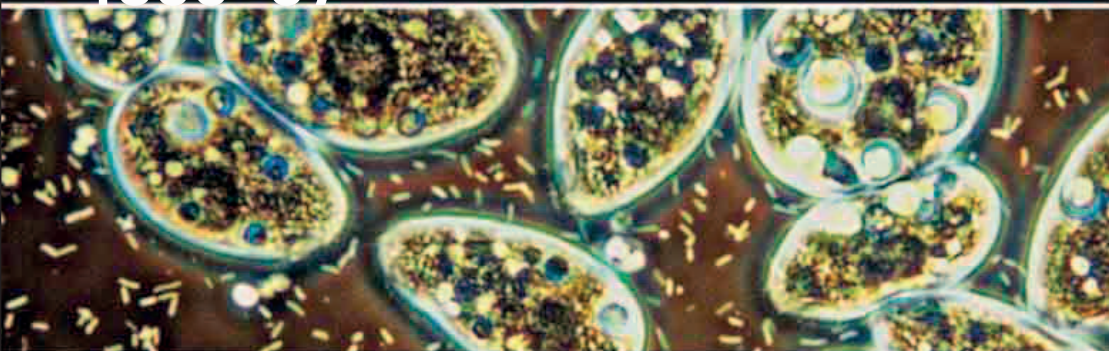




Ernst Haeckel identified one-celled protists as completely separate organisms, although the exact definition is still being debated.

THE INITIAL RELIGIOUS DEBATE OVER DARWIN'S THEORY OF EVOLUTION died down (see 1860), but scientific doubts persisted. One doubt was the question of how traits persist. In 1867, British engineering professor **Fleeming Jenkin** (1833–85) argued that adaptations would eventually blend in with the general population, **getting lost over the generations through what he called a “swamping effect.”**

Unknown to both Darwin and Jenkin, Austrian monk **Gregor Mendel** (1822–84) answered this question with his work on peas (see 1856), which he completed in 1866. Mendel showed that **inherited characteristics**

persist as they are passed on through “factors,” now known as genes. It was not until the early 20th century that scientists fully appreciated the importance of this in relation to evolution.

Another commentator on evolution was German naturalist **Ernst Haeckel** (1834–1919), who argued, wrongly, in 1866 that stages in embryonic development retell evolutionary history. He made drawings to show **similarities between fish and human embryos**. Haeckel also proposed in 1866 that the single-celled organisms called **protists should have a kingdom of their own**, separate from plants and animals.

Also in 1866, German microscopist **Max Schultze** (1825–1974) undertook the most important **early study on the structure of the retina**—the light-sensitive tissue on the inside of the eye. He identified the layers of the retina and drew detailed illustrations of its cellular makeup, including the individual structure of rods and cones—the two types of light-receiving cells inside the back of the eye that react to light and colors (see 1935).

In technology, British engineer **Robert Whitehead** (1823–44) developed the **first self-propelled torpedo** in 1866, a device that later proved devastatingly effective in both World Wars. A year later, Swedish chemist **Alfred Nobel** (1833–96), invented the **explosive dynamite**.

In 1867 Parisian blacksmith **Pierre Michaux** (1813–83) developed one of the **first practical bicycles**, or velocipedes, with pedals and a chain. He also invented one of the **earliest motorcycles**, powered by a tiny steam engine.



Early motorcycle

Dating from between 1867 and 1871, **Pierre Michaux's bicycle** was powered by a small steam engine. It is thought to be the world's first motorcycle.



The Suez Canal was finally opened on November 17, 1869, after 10 years of labor involving tens of thousands of men.

WHILE HE WAS PREPARING A TEXTBOOK

on chemistry in 1868, Russian chemist **Dmitri Mendeleev** (1834–1907) began to wonder if chemical elements could be arranged in a table that **linked their atomic weights and**

properties. Laying out the 60 elements known at the time in weight order, he saw that certain properties were repeated periodically and realized he could organize the elements into eight groups or “periods” of elements. Tellingly, the elements that appear in particular positions in each period have similar properties.

Mendeleev presented his idea, now known as the **Periodic Table**, to a meeting of the Russian Chemical Society on March 6, 1869, and it has been a key reference ever since. What made it so powerful was that

Homo sapien's skull

This European Early Modern Human skull is from Cro-Magnon cave in France. Its discovery provided evidence that humans evolved.



Mendeleev could use it to predict the existence of three as-yet-undiscovered elements to fill gaps

in the table. Over the following 16 years, all three missing elements—gallium, scandium, and germanium—were found, and since then **more than 50 further elements have been identified** and discovered.

Another new element was discovered in 1868 using the knowledge that every substance glows with its own particular spectrum, or light signature (see 1884–85). While studying the spectra of light from the edge of the Sun during a total eclipse, British astronomer **Norman Lockyer** (1836–1920) and French astronomer **Jules Janssen**

17 MILES

THE DISTANCE GIFFARD'S AIRSHIP TRAVELED

1866 Ernst Haeckel identifies protists as separate from plant and animals

1866 Robert Whitehead develops a self-propelling torpedo

March 16, 1867 British surgeon Joseph Lister describes antiseptic surgery

1867 Fleeming Jenkin talks about the problem of blending inheritance with Darwin's theory

May 7, 1867 Alfred Nobel patents dynamite

1867 French engineer Louis-Guillaume Perreux attaches a small steam engine to Michaux's bike to create the first motorcycle

January 20, 1868 British naturalist Charles Darwin publishes his theory of pangenesis

1868 Swedish physicist Anders Ångström maps the solar spectrum

1868 British inventor J. P. Knight invents traffic lights